A MORPHISM INJECTIVE ON POINTS IS SEPARATED

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 4813-4825, following the statement at 4807-4811.

Editorial note. The final sentence defers to 'the argument in the proof of Lemma lemma-diagonal-immersion' , that argument is the one-step closed-local-target reduction (also published); the needed pieces (U -> U x_V U closed immersion via the affine diagonal lemma) are all named.

Allowed standard prerequisites (editorial).

- Affine covering of the fibre product by self fibre products. lemma-fibre-product-affines and lemma-open-fibre-product (same chapter, implicit at ~4819-4821; the covering construction). U x_V U is affine open in X x_Y X.
- Local target criterion for closed immersions. lemma-closed-local-target (same chapter; consumed via the argument of the proof of lemma-diagonal-immersion, card L26408, cited at ~4823-4824). Checks closedness of the diagonal piecewise.

Source reference. Original lemma 01KJ: sources/stacks-schemes/pinned-original-source.tex, source lines 4255–4255. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let $j : X \to Y$ be a morphism of schemes. If j is injective on points, then j is separated.*

Proof. Let z be a point of $X \times_Y X$. Then $x = \mathrm{pr}_1(z)$ and $\mathrm{pr}_2(z)$ are the same because j maps these points to the same point y of Y. Then we can choose an affine open neighbourhood $V \subset Y$ of y and an affine open neighbourhood $U \subset X$ of x with $j(U) \subset V$. Then $z \in U \times_V U \subset X \times_Y X$. Hence $X \times_Y X$ is the union of the affine opens $U \times_V U$. Since $\Delta_{X/Y}^{-1}(U \times_V U) = U$ and since $U \to U \times_V U$ is a closed immersion, we conclude that $\Delta_{X/Y}$ is a closed immersion (see argument in the proof of Lemma 01KJ). $\square$

Date: stacks-project:schemes.tex:local-snapshot-20261012; GNU FDL 1.2 or later.